



Enterprise. Integrated. Connected.

Platform v2.0 — Master Build

"One Platform. Every Business. Every Country. Every Currency."

Founder / CEO	Engola Innocent
Website	www.eic.global
Support	support.eic.global
Developers	developers.eic.global
Version	v2.0 — December 2024
Build Type	Full Production — React + Express + PostgreSQL
Settlement Bank	Centenary Bank Uganda (****8770)

Built by EIC. Owned by the world.

Overview & Vision

EIC (Enterprise. Integrated. Connected.) is the world's most comprehensive, affordable, and intelligent business operating platform. It is a unified SaaS web and mobile application that combines ERP accounting, CRM, IT management, Pi Network services, government tax system integrations, live banking, and mobile money connectivity into one seamlessly beautiful platform.

EIC is designed to decisively outperform every competitor — Zoho, Hyphen AI, QuickBooks, Xero, FreshBooks, SAP, Odoo, NetSuite, and Microsoft Dynamics — in design, affordability, intelligence, and global reach.

● BRAND IDENTITY

Company	EIC — Enterprise. Integrated. Connected.
Tagline	One Platform. Every Business. Every Country. Every Currency.
Domain	www.eic.global
Founder/CEO	Engola Innocent
Primary Color	Deep Space Navy #0A0F2C
Accent Color	Electric Gold #F5A623
Secondary	Emerald Green #00C48C
Dark Mode Base	#06091A
Heading Font	Plus Jakarta Sans (800 weight)
Body Font	Inter Tight (400–700)
UI Style	Glassmorphism — frosted blur, depth shadows

● TECHNICAL ARCHITECTURE

Frontend	React 18 + TypeScript + Vite
Styling	TailwindCSS v4 + inline RGBA glassmorphism
Animation	Framer Motion — 3D tilt cards, entrance reveals
Charts	Recharts — Area, Bar, Line, Pie
Icons	Lucide React
Routing	Wouter (lightweight SPA)

Backend	Node.js + Express 5 + GraphQL (Apollo Server)
Database	PostgreSQL + Drizzle ORM + Redis (cache/sessions)
Auth	NextAuth.js — OAuth (Google/Microsoft/Apple), 2FA, biometric
AI Layer	Anthropic Claude (claude-sonnet-4) via /v1/messages API
Payments	Flutterwave, MTN MoMo API, Airtel Money API, NOWPayments
Infrastructure	AWS (primary) + GCP failover + Cloudflare CDN + DDoS
Security	256-bit AES + TLS 1.3 + GDPR + SOC 2 + PCI DSS Level 1
Mobile	Capacitor.js → native iOS + Android PWA wrappers

Platform Modules

■ Smart Dashboard

MODULE 1

Real-time financial intelligence with 3D-tilting stat cards (CSS perspective transforms).
Revenue, Expenses, and Tax overlaid on a single area chart. Monthly cash flow bar chart.

Key Features

- ♦ KPI cards: Revenue, Net Profit, Tax Due, Active Clients
- ♦ Revenue vs Expenses vs Tax — Area chart (FY 2024)
- ♦ Tax Breakdown Pie — VAT, WHT, PAYE, Income Tax
- ♦ Live transaction ledger with method badges (MTN / Airtel / Bank)
- ♦ Monthly Cash Flow — Inflows vs Outflows bar chart
- ♦ Animated entrance reveals (opacity + translateY)

■ ERP Accounting

MODULE 2

Complete double-entry accounting engine. The backbone of EIC. Production-grade general ledger with drill-down capability, Chart of Accounts builder, and IFRS-compliant export.

Key Features

- ♦ Real-time General Ledger with drill-down
- ♦ P&L; bar chart — Income vs Expenses
- ♦ Accounts Payable & Receivable with aging reports
- ♦ Journal Entry form — post to Centenary / Stanbic / DFCU / Pi Wallet
- ♦ AI-assisted bank reconciliation
- ♦ Export: PDF, CSV, Excel, XBRL, IFRS formats
- ♦ Multi-currency with automatic exchange rate updates
- ♦ Payroll — NSSF/PAYE statutory calculations

■ EFRIS / Tax Compliance

MODULE 3

Mission-critical module. Full integration with Uganda Revenue Authority's Electronic Fiscal Receipting and Invoicing System (EFRIS) via official URA API.

Key Features

- ♦ Direct URA EFRIS portal filing (VAT, WHT, PAYE, Income Tax, Excise Duty)
- ♦ VAT 18% auto-computation on taxable goods/services

- ♦ EFRIS-compliant e-invoices and e-receipts with QR codes
- ♦ TIN validation and lookup
- ♦ URA tax calendar with automated deadline reminders
- ♦ EFRIS credit note and debit note management
- ♦ Offline invoice queuing — auto-syncs when online
- ♦ Full EFRIS compliance audit report generation
- ♦ Kenya KRA iTax, Tanzania TRA, Rwanda RRA, Nigeria FIRS, SARS, HMRC MTD

■ Pi Wallet — Mobile Money

MODULE 4

Unified mobile money wallet. Handles MTN MoMo, Airtel Money, M-Pesa, and direct bank transfers. All payments auto-settle to Centenary Bank account.

Key Features

- ♦ Pi Wallet balance display + transaction history
- ♦ Send via MTN MoMo, Airtel Money, M-Pesa, Bank Transfer
- ♦ Auto-settlement → Centenary Bank (****8770)
- ♦ Real-time balance sync for MTN and Airtel wallets
- ♦ Payment request links — shareable via WhatsApp/SMS/email
- ♦ Bulk payment disbursement (payroll, supplier payments)
- ♦ Standing orders and scheduled payment automation

Platform Modules (cont.)

■ Banking Integration

MODULE 5

Connected accounts across Centenary Bank, Stanbic, DFCU, and international banks. Real-time balance sync via direct bank APIs and Open Banking standards.

Key Features

- ♦ Centenary Business · Stanbic Operating · DFCU Savings — live cards
- ♦ Fund transfer form with confirmation flow
- ♦ Bank statement download (PDF, CSV, OFX, MT940)
- ♦ AI-assisted automatic transaction categorization
- ♦ Two-way bank reconciliation with accounting ledger
- ♦ Plaid API (USA/UK/EU) · Mono (Nigeria/Ghana/Kenya) · Stitch (South Africa)
- ♦ International: Open Banking PSD2, Stripe Financial Connections

■ CRM

MODULE 6

Full customer relationship management with AI-powered lead scoring, visual Kanban pipeline, and multi-country contact management.

Key Features

- ♦ Contacts table — AI lead scores (0–100), lifetime value, status
- ♦ Kanban pipeline: Lead → Qualified → Proposal → Negotiation → Closed
- ♦ Country revenue analytics bar chart
- ♦ Add/edit contacts with country + pipeline stage selection
- ♦ WhatsApp Business API integration for customer messaging
- ♦ Email & SMS campaign builder with A/B testing
- ♦ Customer portal — clients view invoices, make payments
- ♦ CSAT and NPS survey tools

■ IT Management

MODULE 7

Hardware and software asset register with QR code tagging. Security scoring, network monitoring, and internal IT helpdesk fully integrated with the ERP accounting module.

Key Features

- ♦ Asset register — hardware & software with tag, dept, expiry

- ♦ Security health scoring — Firewall, Antivirus, Patches, Access Control, Backup
- ♦ License tracking with expiry alerts
- ♦ Network monitoring: uptime, latency, bandwidth, device status
- ♦ IT budget & spend linked to ERP accounting module
- ♦ User provisioning workflows tied to HR onboarding
- ♦ Cloud storage integration: Google Drive, OneDrive, Dropbox, Box

π Crypto & Pi Network

MODULE 8

Native Pi Network wallet. Accepts Pi Coin as first-class payment (20% discount on subscriptions). Full crypto payment gateway via Flutterwave/NOWPayments.

Key Features

- ♦ Live crypto rates: BTC, ETH, USDT, SOL, BNB
- ♦ Crypto-to-UGX real-time conversion display
- ♦ 20% subscription discount for Pi Coin payments
- ♦ Generate crypto payment links for invoices
- ♦ Pi Coin, Bitcoin, Ethereum, USDT, USDC, BNB, Solana accepted
- ♦ Fiat on/off ramp for supported countries
- ♦ All crypto transactions linked to accounting ledger entries

■ ARIA — AI Intelligence Layer

MODULE 9

Context-aware AI assistant powered by Anthropic's Claude model. Embedded throughout every module. Answers questions about your own financial data in natural language.

Key Features

- ♦ Live Claude-powered chat via Anthropic /v1/messages API
- ♦ Full financial context pre-loaded (revenue, expenses, tax, clients)
- ♦ Smart invoice generation from natural language
- ♦ Automatic expense categorization from bank feeds
- ♦ Anomaly detection — unusual transactions, duplicate payments
- ♦ 12-month revenue and cash flow forecasting
- ♦ CRM churn prediction with intervention recommendations
- ♦ EFRIS tax filing deadline reminders and pre-filled summaries
- ♦ Multilingual document translation (100+ languages)

Payment & Settlement Architecture

■■ SECURITY NOTICE: Bank account credentials must be stored ONLY in your backend .env file and never exposed in frontend source code. The frontend displays masked values (****8770) and communicates via your API server.

● SETTLEMENT ACCOUNT

Bank Name	Centenary Bank Uganda
Account Name	EIC Enterprise Ltd
Account Number	Stored in backend .env as CENTENARY_ACCOUNT
Branch	Kampala Main
Currency	UGX (primary) — multi-currency supported
Display in UI	****8770 (masked for security)

● PAYMENT GATEWAY STACK

MTN MoMo	MTN Mobile Money API — Collections + Disbursements + Transfers
Airtel Money	Airtel Money API — Uganda, Kenya, Tanzania, Rwanda, Zambia
M-Pesa	Safaricom Daraja API — Kenya, Tanzania, Mozambique, DRC, Ghana
Flutterwave	Pan-Africa + global payment infrastructure + settlement
Paystack	Nigeria, Ghana, Kenya, South Africa
Stripe	USA, UK, EU — card processing + Financial Connections
NOWPayments	Crypto gateway — BTC, ETH, USDT, USDC, BNB, SOL, Pi Coin
Pi Network	Pi SDK — 20% subscription discount, direct Pi Coin acceptance
Plaid	USA/Canada/UK/EU bank account linking + transaction sync

● BACKEND ENVIRONMENT VARIABLES (.ENV)


```

# ■■ Centenary Bank Settlement ■■
CENTENARY_ACCOUNT=32046608770
CENTENARY_NAME=EIC Enterprise Ltd
CENTENARY_BANK=Centenary Bank Uganda

# ■■ Mobile Money APIs ■■
MTN_MOMO_SUBSCRIPTION_KEY=your_subscription_key
MTN_MOMO_API_USER=your_api_user_uuid
MTN_MOMO_API_KEY=your_api_key
MTN_MOMO_ENV=production # sandbox | production

AIRTEL_CLIENT_ID=your_airtel_client_id
AIRTEL_CLIENT_SECRET=your_airtel_secret
AIRTEL_ENV=production

MPESA_CONSUMER_KEY=your_daraja_consumer_key
MPESA_CONSUMER_SECRET=your_daraja_secret
MPESA_SHORTCODE=your_till_or_paybill

# ■■ Payment Gateways ■■
FLW_SECRET_KEY=FLWSECK-xxxxxxxxxxxxx
FLW_PUBLIC_KEY=FLWPUBK-xxxxxxxxxxxxx
PAYSTACK_SECRET_KEY=sk_livexxxxxxxxxxxxx
STRIPE_SECRET_KEY=sk_livexxxxxxxxxxxxx

# ■■ Crypto ■■
NOWPAYMENTS_API_KEY=your_nowpayments_key
PI_APP_IDENTIFIER=your_pi_app_id
PI_API_KEY=your_pi_api_key

# ■■ AI ■■
ANTHROPIC_API_KEY=sk-ant-xxxxxxxxxxxxx

# ■■ Database ■■
DATABASE_URL=postgresql://user:pass@host:5432/eic_db
REDIS_URL=redis://localhost:6379

# ■■ Auth ■■
NEXTAUTH_SECRET=your_nextauth_secret_min_32chars
GOOGLE_CLIENT_ID=your_google_oauth_client_id
GOOGLE_CLIENT_SECRET=your_google_oauth_secret

```

■ **LIVE MONEY FLOW:** To route real payments to your Centenary account, configure Flutterwave's settlement settings in their dashboard to auto-settle all collections to account 32046608770 at Centenary Bank Uganda. MTN MoMo and Airtel Money disbursements are configured via their respective API environments.

UI Design System

COLOR PALETTE

Deep Space Navy	#0A0F2C	Page background, primary surfaces
Dark Navy	#06091A	Sidebar, header background
Electric Gold	#F5A623	Primary accent, CTA buttons, active states
Gold Light	#FFBE55	Hover states, gradients
Emerald Green	#00C48C	Success states, EFRIS compliance, positive values
Soft White	#F8F9FF	Primary text
Muted Blue	#8090B8	Secondary text, labels
MTN Yellow	rgba(255,204,0,0.9)	MTN MoMo payment tint — payment UI only
Airtel Red	rgba(255,60,60,0.9)	Airtel Money tint — payment UI only
M-Pesa Green	rgba(0,160,60,0.9)	M-Pesa tint — payment UI only
Bank Blue	rgba(79,195,247,0.9)	Bank transfer tint — banking UI only
Error Red	#FF6B6B	Error states only

GLASSMORPHISM LAYERS

Card Background	rgba(255,255,255, 0.04) + backdrop-filter: blur(12px)
Card Border	rgba(255,255,255, 0.10)
Gold Card Border	rgba(245,166,35, 0.30)
Hover Background	rgba(255,255,255, 0.07)
Input Background	rgba(255,255,255, 0.05)
Input Border	rgba(255,255,255, 0.10)
Active Nav Item	rgba(245,166,35, 0.12) + border rgba(245,166,35, 0.22)
3D Tilt Cards	perspective(600px) rotateX/Y on mouse move

COMPONENT SPECS

Primary Button	height:42 · background:gold · color:navy · borderRadius:10 · font-weight:700
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Ghost Button	height:36 · border:1px rgba(255,255,255,0.10) · color:muted
Input Field	height:44 · px:14 · rounded:10 · bg:rgba(255,255,255,0.05)
Card Radius	borderRadius: 16px
Badge / Pill	borderRadius:999 · font:11px bold · uppercase · colored bg
Section Badge	border:1px rgba(245,166,35,0.22) · gold text · uppercase tracking
Sidebar Width	228px expanded · 60px collapsed · collapsible toggle
Header Height	54px · live EAT clock · notification bell · user avatar

Code Structure & Key Snippets

● FILE STRUCTURE

```
eic-platform/
├── frontend/
│   ├── src/
│   │   ├── App.jsx # Root – sidebar + header + view router
│   │   ├── views/
│   │   │   ├── Dashboard.jsx # Smart Dashboard – charts + KPI cards
│   │   │   ├── Accounting.jsx # ERP ledger, P&L; , journal entry
│   │   │   ├── EFRIS.jsx # URA EFRIS filing + submission history
│   │   │   ├── Wallet.jsx # Pi Wallet + MTN / Airtel / M-Pesa
│   │   │   ├── Banking.jsx # Centenary / Stanbic / DFCU accounts
│   │   │   ├── CRM.jsx # Contacts, pipeline Kanban, analytics
│   │   │   ├── IT.jsx # Asset register + security scoring
│   │   │   ├── Crypto.jsx # Pi Network + BTC/ETH/USDT/SOL
│   │   │   ├── AI.jsx # ARIA – Claude-powered chat
│   │   ├── components/
│   │   │   ├── GlassCard.jsx # Glassmorphism card + hover lift
│   │   │   ├── StatCard3D.jsx # 3D tilt stat card
│   │   │   ├── GoldBtn.jsx # Primary gold CTA button + pulse glow
│   │   │   ├── ChartTooltip.jsx # Branded recharts tooltip
│   │   ├── brand.js # Design tokens / color constants
│   │   └── vite.config.ts
│   └── backend/
│       ├── src/
│       │   ├── routes/
│       │   │   ├── payments.js # MTN MoMo, Airtel, Flutterwave handlers
│       │   │   ├── efris.js # URA EFRIS API proxy
│       │   │   ├── banking.js # Centenary + Stanbic + DFCU API
│       │   │   ├── ai.js # Anthropic Claude API proxy
│       │   ├── models/ # Drizzle ORM schemas
│       │   ├── middleware/
│       │   │   ├── auth.js # JWT + bcrypt
│       │   │   ├── security.js # Rate limit, CSRF, XSS sanitize
│       │   └── server.js # Express 5 entry point
│       ├── .env # All secrets (NEVER commit to git)
│       └── docker-compose.yml
```

● ARIA — AI CHAT INTEGRATION (CORE SNIPPET)

```
// frontend/src/views/AI.jsx – ARIA chat handler
const send = async () => {
  const apiMsgs = msgs.slice(1).map(m=>({role:m.role, content:m.content}));
  const res = await fetch('https://api.anthropic.com/v1/messages', {
    method: 'POST',
    headers: { 'Content-Type': 'application/json' },
    body: JSON.stringify({
      model: 'claude-sonnet-4-20250514',
      max_tokens: 1000,
      system: `You are ARIA, EIC's AI layer. User: Engola Innocent.`
```

```

Revenue: UGX 48.2M, Tax Due: UGX 4.1M, Clients: 284.
Settlement: Centenary Bank Uganda.`
messages: apiMsgs
})
});
const data = await res.json();
const reply = data.content?.find(b=>b.type==='text')?.text;
setMsgs([...msgs, {role:'assistant', content:reply}]);
};

```

● MTN MOMO PAYMENT HANDLER (BACKEND)

```

// backend/src/routes/payments.js
const { MTN_MOMO_SUBSCRIPTION_KEY, CENTENARY_ACCOUNT } = process.env;

router.post('/momo/request-to-pay', async (req, res) => {
  const { amount, phone, currency = 'UGX', reference } = req.body;
  const response = await fetch(
    'https://proxy.momoapi.mtn.com/collection/v1_0/requesttopay',
    {
      method: 'POST',
      headers: {
        'Authorization': `Bearer ${await getMomoToken()}`,
        'X-Reference-Id': reference,
        'X-Target-Environment': 'production',
        'Ocp-Apim-Subscription-Key': MTN_MOMO_SUBSCRIPTION_KEY,
        'Content-Type': 'application/json',
      },
      body: JSON.stringify({
        amount, currency, externalId: reference,
        payer: { partyIdType: 'MSISDN', partyId: phone },
        payerMessage: 'EIC Platform Payment',
        payeeNote: `Settlement: ${CENTENARY_ACCOUNT}`,
      })
    }
  );
  // Settlement auto-routes to Centenary Bank via Flutterwave
  res.json({ status: response.status === 202 ? 'accepted' : 'failed' });
});

```

■ To go live: (1) Register on MTN MoMo Developer Portal and get production keys. (2) Set up Flutterwave merchant account — configure settlement to Centenary Bank account in Flutterwave dashboard. (3) All mobile money collections auto-settle to your Centenary account within 24–48 hours.

Pre-Launch Quality Checklist

All items below must be verified before publishing the EIC application to production.

● PAYMENTS & BANKING

■	All bank connections tested in production with real accounts (not just sandbox)
■	MTN MoMo, Airtel Money, M-Pesa APIs tested with live transactions
■	Flutterwave settlement configured → Centenary Bank (****8770)
■	All currency conversions verified with live exchange rate feeds
■	Payment gateways tested end-to-end (charge, refund, dispute)
■	PCI DSS Level 1 compliance verified for all payment processing

● TAX & COMPLIANCE

■	URA EFRIS: test invoice submission, QR generation, VAT return filing
■	ASYCUDA integration: customs declaration tested for 3+ countries
■	Kenya KRA iTax, Tanzania TRA, Rwanda RRA integrations tested
■	All tax system integrations verified against government portals
■	EFRIS offline queuing tested — invoice created offline, syncs when reconnected

● AI & PLATFORM

■	ARIA (Claude) responds correctly to financial queries with real data
■	All 9 modules functional with no runtime errors
■	Pi Coin 20% discount applied correctly at checkout
■	ARIA context includes up-to-date revenue, expense, client data
■	All AI features tested with real business data scenarios

● SECURITY & PERFORMANCE

■	Full penetration test and security audit completed
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- GDPR consent flows and data deletion requests verified
- Load testing: 10,000 concurrent users without performance degradation
- Cross-browser testing: Chrome, Safari, Firefox, Edge, Opera Mini (Africa)
- Mobile apps (iOS + Android) tested on 10+ device types
- Accessibility audit passed (WCAG 2.1 AA)
- www.eic.global domain live, SSL active, all subdomains resolving

● SIGN-OFF

- CEO profile page reviewed and approved by ENGOLA INNOCENT before go-live
- All 100+ language switches tested including RTL layouts (Arabic, Hebrew, Urdu)
- PWA offline mode tested: create invoice offline, sync when reconnected
- Final QA sign-off from all department leads

Final Vision Statement

"EIC is not just software. It is the financial nervous system of every business on earth."

Built by ENGOLA INNOCENT with a singular mission: to give every entrepreneur, every SME, every corporation — regardless of their country, language, or currency — the same world-class tools that were previously only available to Fortune 500 companies.

EIC ends the era of fragmented, expensive, and exclusionary business software.

**One Platform. Every Business. Every Country. Every Currency.
Forever.**

www.eic.global | support.eic.global | developers.eic.global

Built by EIC. Owned by the world.